

Q2. Subscription Billing System

A streaming platform offers two subscription plans—**Basic** and **Premium**—each with different billing rules. The system should calculate the **final payable amount** for each subscription.

Design Requirement

Implement the solution using **Object-Oriented Programming (OOP)**.

- Create a **base class** Subscription
- Derive the following subclasses:
 - BasicPlan
 - PremiumPlan
- Each class should implement a method **generate_invoice()** to compute the final payable amount.

Subscription Details

Each subscription contains:

- Subscriber ID
- Subscriber Name
- Monthly Fee
- Number of Months

Billing Rules

Basic Plan

- $\text{Total} = \text{monthly_fee} \times \text{months}$
- If months > 6 → apply **5% discount**

Premium Plan

- $\text{Total} = \text{monthly_fee} \times \text{months}$
- If months > 6 → apply **10% discount**

- Add ₹200 service charge

Constraints

- $1 \leq \text{months} \leq 60$
- $1 \leq \text{monthly_fee} \leq 1,000,000$
- Subscriber ID must be positive

Sample Input

2

Basic 101 Riya 1000 8

Premium 102 Arjun 2000 5

Sample Output

7600.0

10200